TEXORA · LECTURE SHEET

The Comber

The machine that upgrades cotton from carded to combed: it combs the clamped fringe of a lap with a rotating needle cylinder, pulls the long fibres forward through a top comb, and rejects the short fibres, neps and trash as noil — buying yarn quality with raw material.

TE-2201 · SPINNING TECHNOLOGY-II · TOPIC 2 · LEVEL 2 TERM 2

1. Objects of the comber

- To remove **short fibres** (below a chosen effective length) from the lap — collected as waste called **noil**.
- To remove remaining **neps, seed fragments and trash** that carding left behind.
- To **straighten and parallelise** the retained long fibres far beyond draw-frame standards, removing fibre hooks.
- To deliver a **clean, even sliver** from which stronger, finer, smoother and more lustrous yarns can be spun — combed yarns command counts and qualities that carded yarns cannot reach.

The price is deliberate: the noil is typically 10–25 % of the feed. Combing is therefore used where the yarn's quality justifies it — fine counts, shirtings, sewing threads, high-grade knitwear.

2. The working parts in one view

The modern machine is a **rectilinear comber** with (usually) eight heads working in parallel. At each head: the **lap** unwinds over the feed stand; a grooved **feed roller** advances the lap in small steps; the **nipper assembly** — a lower nipper plate and an upper nipper knife carried on a swinging frame — clamps the lap and presents its fringe first to the **combing cylinder** (a half-lap shaft clothed over part of its circumference with rows of fine needles or saw-tooth wire) and then to the pair of **detaching rollers**. A **top comb**, a single row of needles on a blade, drops into the fringe during detachment. A rotating **brush** cleans the cylinder, and an air stream carries the stripped noil away. The detached web passes over a polished detaching table, is condensed through a trumpet, drafted lightly by draw-box rollers at the machine's delivery side (all eight heads are combined and drafted together), and coiled into a can.

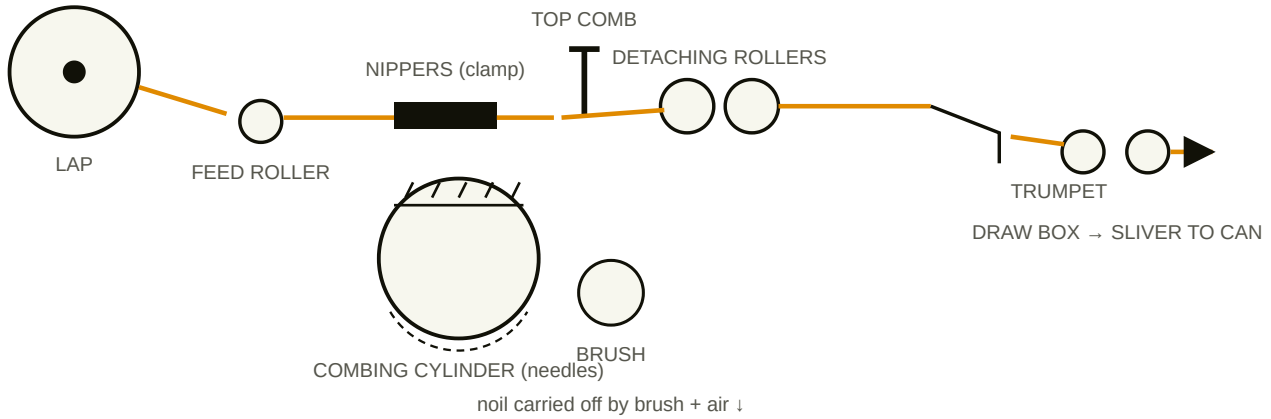
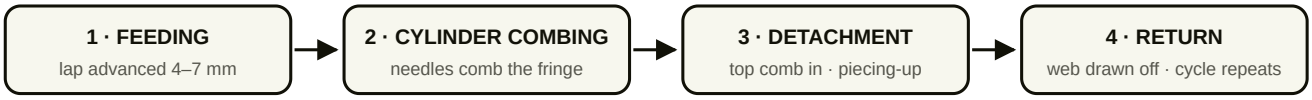


Figure 1 — One combing head, side view. The nipper assembly swings the clamped fringe down to the cylinder for combing, then forward to the detaching rollers for piecing-up; the top comb drops in only during detachment.

3. The combing cycle, step by step

One revolution of the combing cylinder is one cycle — one **nip** — repeated 300–600 times a minute. All motions are timed from the cylinder shaft against a graduated disc, the **index wheel**, divided into 40 equal parts. The four phases:

1. **Feeding.** The nippers are closed on the lap and stand back from the detaching rollers. The feed roller turns and pushes the lap forward by the set **feed length** (roughly 4–7 mm), so a fresh fringe projects from the nipper mouth. (During this phase the detaching rollers turn backwards briefly, pushing the tail of the previously formed web back toward the nippers, ready for piecing-up.)
2. **Cylinder combing.** The nipper assembly swings forward and downward, bringing the fringe close over the cylinder while the top nipper lifts slightly at the right instant so the fringe lies across the cylinder's path. The needle rows sweep through the fringe from root to tip: long fibres are held by the nippers and combed straight; short fibres, neps and trash, not held by the nipper, are carried away on the needles. The rotating brush strips the cylinder, and suction air removes the noil.
3. **Detachment and piecing-up.** The nipper assembly swings forward toward the detaching rollers, which grip the combed fringe and lay it onto the returning tail of the previous web — the overlap is the **piecing-up**. As the detaching rollers now rotate forward they pull the fringe out of the lap. At this moment the **top comb** descends and penetrates the fringe: the detaching rollers draw the long fibres through its needles, so fibres too short to be gripped, and the neps they carry, are held back and left in the fringe residue — to be removed by the cylinder in the next cycle. Detachment is the second combing action of the cycle, and the only one that cleans the *tail* ends of the fibres.
4. **Return.** The top comb rises, the nippers close on the lap again and swing back; the detached web is drawn off across the table, condensed into sliver, and the cycle repeats.



ONE NIP = ONE CYLINDER REVOLUTION · TIMED ON THE INDEX WHEEL (40 DIVISIONS)

Figure 2 — The combing cycle. Feeding and cylinder combing prepare the fringe; detachment with the top comb in place forms the web and cleans the fibre tails.

4. The settings that decide what is removed

4.1 Detachment setting

The **detachment setting** is the distance between the nipper bite point and the nip of the detaching rollers at their closest approach — roughly 17–27 mm depending on staple length and the noil target. It fixes the shortest fibre that the detaching rollers can reach and pull out of the fringe. Only fibres long enough to bridge the gap between the nipper bite and the detaching-roller nip can be seized and drawn off; every shorter fibre stays behind in the fringe and is removed as noil in the next combing. So a **wider (larger) setting** means fewer fibres can bridge the gap: noil rises, and the sliver is cleaner, but yield drops. A **closer (smaller) setting** takes more of the fringe: noil falls, but more short fibre passes into the sliver and evenness suffers. The detachment setting is therefore the master control of noil percentage, chosen together with the cotton's staple length: too close a setting on a long-staple cotton also tears fibre and raises neps.

4.2 Top comb penetration

The depth to which the top comb's needles enter the fringe during detachment — typically adjustable over about 0.5 mm steps within roughly −0.5 to +1.0 mm of the fringe plane. **Deeper penetration** holds back more short fibre and nep at the tail end: sliver cleaner, noil slightly higher, but excessive depth obstructs the passage of good fibre, raises fibre breakage and can mark the web. Setting is a compromise verified by testing the sliver and the noil.

4.3 Feed length and feeding system

FEATURE	FORWARD FEEDING	BACKWARD FEEDING
When the lap is fed	While the nippers are swinging *away* from the detaching rollers (fringe combed after feeding, before detachment)	While the nippers are swinging *toward* the detaching rollers

Typical feed length	Shorter, ≈ 4.3–5.3 mm	Longer, ≈ 5.5–7.0 mm
Combing intensity	Moderate — the part of the fringe fed this cycle is detached in the same cycle	Higher — the fed fringe is combed by the cylinder once more before detachment, so each fibre is worked over a longer effective length
Noil at equal settings	Lower	Higher (more short fibre worked loose)
Best used for	Medium counts, production emphasis	Fine counts, quality emphasis

4.4 Other influencing parameters

- **Lap weight and evenness** — a heavier lap feeds more mass per nip but overloads the needles and thickens the fringe; an uneven lap defeats every setting (see Topic 1).
- **Cylinder speed (nips/min)** — raises production but shortens combing time per fringe and can increase fibre damage and nep generation if pushed.
- **Needle condition** — bent, hooked or missing needles pass uncombed bundles into the sliver ("fish-hook" defects) and raise noil irregularly.
- **Timing (index wheel settings)** — feed, nipper opening, top comb entry and detaching-roller reversal must occur at their designed index numbers; mistiming shows up as torn fringes, poor piecing and steps in the web.
- **Brush and suction setting** — a badly cleaned cylinder re-deposits noil into the fringe.

5. Noil — the waste that is the point

Comber waste (noil) is everything the machine rejects: predominantly short fibres, with neps, seed coat fragments and trash, plus — unavoidably — a small proportion of spinnable fibre that was simply too short to bridge the detachment gap. **Waste control** means holding the noil percentage at its target (set by the quality the yarn must reach and the price the cotton can bear): it is checked by regularly weighing feed lap and delivered sliver, or directly by collecting and weighing the noil over a timed run. Noil that creeps upward wastes money; noil that creeps downward silently downgrades every yarn spun from the sliver.

Effect of noil percentage on yarn quality: raising the noil from, say, 10 % to 18 % removes progressively more short fibre, so the yarn becomes measurably stronger (load is carried by long fibres), more even (short fibres are a main cause of thick/thin places), cleaner and smoother, with fewer imperfections and a better appearance after dyeing — and finer counts become spinnable at all. Beyond

the useful range the gains flatten while the cost keeps rising, because each extra percent of noil is increasingly made of usable fibre. The correct noil is an economic optimum, not a maximum.

6. Drafting and coiling at delivery

The eight webs are laid together and passed through the **draw box**, a small roller drafting system (draft roughly 8–15 across the combined webs, depending on head count and target sliver weight) which levels the periodic irregularity that piecing-up inevitably leaves at each joint. The sliver then passes over the delivery table to a **coiler**: a rotating trumpet and coiler plate lay the sliver in controlled overlapping coils inside a spring-loaded can — the **sliver coiling system** — so that a full can holds maximum length without tangles and the sliver can later be withdrawn cleanly, from the top, without stretching.

WORKED EXAMPLE — COMBER PRODUCTION

An 8-head comber runs at 350 nips/min, feed length 5 mm per nip, lap weight 64 g/m, efficiency 90 %, noil 15 %.

Feed per nip per head = (5 ÷ 1000) m × 64 g/m = 0.32 g.

Lap consumed = 0.32 g × 350 nips/min × 8 heads × 60 min = 53.76 kg/h theoretical; × 0.90 = **48.4 kg/h** actual feed.

Sliver delivered = 48.4 × (1 − 0.15) ≈ **41.1 kg/h**. The remaining ≈ 7.3 kg/h leaves as noil — the visible cost of the quality the machine exists to give.

7. Effect of setting changes on sliver quality — summary

CHANGE	NOIL %	SLIVER / YARN EFFECT
Detachment setting wider	Rises	Cleaner, more even sliver; yield and production of sliver fall
Detachment setting closer	Falls	More short fibre retained; evenness and strength fall
Top comb deeper	Rises slightly	Fewer neps and short fibres at tail ends; too deep → fibre breakage
Feed length longer	Rises slightly	More material per nip (production up); fringe thicker, combing less thorough
Cylinder speed higher	About constant	Production up; excessive speed → fibre damage, nep increase

Change to backward feeding	Rises	More intensive combing; finer, cleaner sliver at lower yield

Glossary

Noil — the waste extracted by the comber: short fibres, neps and trash. **Nip** — one combing cycle; also the grip point between rollers. **Nippers** — the clamping jaws that hold the lap fringe. **Fringe** — the length of lap projecting from the nippers to be combed and detached. **Detachment setting** — the gap between nipper bite and detaching-roller nip at closest approach. **Piecing-up** — overlapping the new detached fringe onto the tail of the previous web so the web is continuous. **Index wheel** — the 40-division timing disc on the cylinder shaft against which all cyclic motions are set. **Top comb** — the fixed needle blade that enters the fringe during detachment. **Draw box** — the delivery drafting system combining all heads' webs into one sliver.

Key points

• The comber removes short fibre, neps and trash as noil, and delivers a parallelised, clean sliver for fine, strong yarns.
• One cycle = feeding → cylinder combing → detachment with the top comb in (piecing-up) → return; timed on the 40-division index wheel.
• The detachment setting is the master control of noil %; top comb depth cleans the fibre tails; backward feeding combs more intensively than forward feeding.
• More noil = stronger, evener, cleaner yarn — up to an economic optimum; beyond it, good fibre is thrown away.
• Lap quality, needle condition and timing decide whether the settings can do their work.
• Production follows from nips/min × feed length × lap weight × heads, less efficiency and less the noil removed.

Practice

1. A comber with 8 heads runs at 400 nips/min with a feed of 4.7 mm on laps of 68 g/m at 92 % efficiency, extracting 16 % noil. Calculate the sliver production in kg/h.
2. Put the four phases of the combing cycle in order and state, for each, which machine element does the work.
3. A mill wants to spin a finer count from the same cotton. Which two settings would you change first, in which direction, and what will each cost?

4. Why does detachment need the top comb, when the cylinder has already combed the fringe? Which end of the fibre does each action clean?
5. Noil from a comber rises from 14 % to 19 % over a week with no setting change. Give four machine-side causes to inspect.
6. Explain why combed sliver still passes through a draw box and (usually) further draw-frame passages although the comber has "finished" cleaning the fibre.

Answer key

1. Per nip per head: (4.7 ÷ 1000) × 68 = 0.3196 g. Feed = 0.3196 × 400 × 8 × 60 = 61.36 kg/h; × 0.92 = 56.5 kg/h fed. Sliver = 56.5 × 0.84 ≈ **47.4 kg/h**.
2. Feeding (feed roller) → cylinder combing (combing cylinder against the clamped fringe) → detachment and piecing-up (detaching rollers, with the top comb penetrating the fringe) → return (nippers close and swing back; web drawn off through the draw box).
3. (i) Widen the detachment setting — raises noil, removes more short fibre; cost: yield and sliver production fall. (ii) Deepen the top comb penetration (or change to backward feeding) — cleaner fibre tails / more intensive combing; cost: more noil again and some risk of fibre breakage if overdone.
4. The cylinder combs only the fringe that projects from the nippers — the leading portion of each fibre; the tail ends remain inside the lap, uncombed. During detachment the long fibres are drawn *through* the stationary top comb, which combs their tails and holds back short fibre and neps at that end. Both ends of every fibre are thus combed once per cycle.
5. Any four: needles damaged or hooked sections worn (fibre no longer held/released correctly); detachment setting drifted wider after maintenance; top comb set deeper or damaged; lap weight increased (thicker fringe → more left behind); feed length changed longer; brush/suction fault recirculating waste; nipper knife or plate worn so the clamp slips and fringe length varies.
6. Combing cleans and parallels, but piecing-up joins the web in discrete tufts, leaving periodic thick joints, and the fibre arrangement in the web is not yet a controlled sliver. The draw box and later draw frames draft out that periodicity, double the strands to average remaining irregularities, and set the sliver to the exact weight the next process requires.
